Abhigyaan Singh

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Professional Summary

B.Tech Information Technology student with strong foundations in C++, data structures and algorithms, and Object Oriented Programming. Experienced in building modular C++ applications, optimizing collision detection algorithms, and implementing multithreaded processing. Familiar with performance benchmarking, automated testing, and developing interactive applications using OpenGL and CMake.

Relevant Coursework

Data Structures and Algorithms (DSA), Operating Systems (OS), Relational Database Management Systems (RDBMS), CyberSecurity, Deep Learning, Generative AI (GenAI), Computer Networks, Embedded Systems, Digital Systems and Computer Organisation, Information Security, Data Mining, Full Stack Web Development, Blockchain Technology, Cloud Computing

Work Experience

Vodafone Idea | IT Intern Jun 2026 – Jul 2026

- Processed and analyzed ~6 million telecom subscriber and retailer records using Impala SQL for churn and activation analysis
- Implemented RANK analysis, dynamic date filtering, partitioning, and monthly cohort analysis for subscriber insights.
- Created and validated SQL tables and data-processing workflows to support large-scale subscriber data processing.

Projects

Teleported | React.js | Node.js | Express.js | MongoDB (repo-link)

- Developed a full-stack travel platform using the MVC design pattern with a responsive React.js interface, reusable UI components, and location-based features.
- Implemented client-side routing, state management, form validation, and search/filtering to support interactive user workflows.
- Built React-based UI components and integrated RESTful APIs with JWT-based authentication, password hashing, CRUD operations, input validation, and structured error handling.
- Designed MongoDB data models with relationships and external Maps API integration for location features.
- Implemented unit tests using Jest to validate application logic, invalid inputs, and edge cases and used AI tools to generate test scenarios.

Hisaab | Flutter | Dart | SQLite | Android (repo-link)

- Developed an Android expense tracker using Flutter and Dart to automatically extract financial transactions from bank SMS messages.
- Implemented SMS permissions and regex-based parsing to extract transaction details from incoming messages.
- Built transaction filtering and validation logic to identify relevant debit/credit messages and reduce incorrect transaction extraction.
- Built SQLite-based transaction storage, monthly expense aggregation, and interactive pie charts and graphs to visualize spending patterns and category-wise expenses.

Traffic Congestion Analysis – Pandas | NumPy | Scikit-learn | XGBoost | LightGBM | Optuna | SHAP (repo-link)

- Integrated traffic and weather data, applying preprocessing and feature engineering to build predictive models for congestion analysis.
- Engineered cyclic time, weather, lag, rolling, and junction-level features to capture congestion patterns.
- Compared Random Forest, XGBoost, and LightGBM, addressing class imbalance with SMOTE and optimizing the best-performing XGBoost model using 40-trial Optuna tuning.
- Achieved 96.98% test accuracy, 0.9944 ROC-AUC, and 0.939 F1-score for congestion prediction, using threshold optimization, TimeSeriesSplit validation, probability calibration, and SHAP-based feature interpretation.

Technical Skills

Languages: C/C++, Java, SQL
Concepts: Data Structures, Algorithms, Object Oriented Programming, RDBMS, Operating Systems
Frontend: React.js, HTML5, CSS3, Flutter, Dart
Backend and Web: Express.js, Node.js, REST APIs
Databases: MySQL, MongoDB, Impala SQL, PL/SQL
Tools: Git, GitHub

Education

Manipal Institute of Technology, Manipal | 2023–2027
B.Tech Information Technology
CGPA - **8.18/10** (as of semester VI)
Loyola Public School, Lucknow- Class XII | 2022
Grade:95%
Rani laxmi Bai Memorial School, Lucknow - Class X | 2020
Grade : 94%

Certifications

- Meta Front-end Developer Professional Course
- Google AI Professional Course